Example 3.41 If $A+B+C=\pi$, prove that $\cos^2 A+\cos^2 B+\cos^2 C=1-2\cos A\cos B\cos C$.

Solution:

$$\begin{aligned}
\cos^2 A+\cos^2 B+\cos^2 C &= \frac{1}{2}\left[2\cos^2 A+2\cos^2 B+2\cos^2 C\right]\\
&= \frac{1}{2}\left[(1+\cos 2A)+(1+\cos 2B)+(1+\cos 2C)\right]\\
&= \frac{3}{2}+\frac{1}{2}\left[(\cos 2A+\cos 2B)+\cos 2C\right]\\
&= \frac{3}{2}+\frac{1}{2}\left[2\cos(A+B)\cos(A-B)+(2\cos^2 C-1)\right]\\
&= \frac{3}{2}+\frac{1}{2}\left[-2\cos C\cos(A-B)+2\cos^2 C-1\right] \quad (A+B=\pi-C)\\
&= \frac{3}{2}-\frac{1}{2}+\frac{1}{2}\left[-2\cos C\left(\cos(A-B)-\cos C\right)\right]\\
&= 1-\cos C\left[\cos(A-B)-\cos C\right]\\
&= 1-\cos C\left[\cos(A-B)+\cos(A+B)\right]\\
&= 1-\cos C\left[2\cos A\cos B\right]\\
&= 1-2\cos A\cos B\cos C
\end{aligned}$$

Exercise - 3.7

1. If $A+B+C=180^\circ$, prove that
   (i) $\sin 2A+\sin 2B+\sin 2C=4\sin A\sin B\sin C$
   (ii) $\cos A+\cos B-\cos C=-1+4\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$
   (iii) $\sin^2 A+\sin^2 B+\sin^2 C=2+2\cos A\cos B\cos C$
   (iv) $\sin^2 A+\sin^2 B-\sin^2 C=2\sin A\sin B\cos C$
   (v) $\tan\frac{A}{2}\tan\frac{B}{2}+\tan\frac{B}{2}\tan\frac{C}{2}+\tan\frac{C}{2}\tan\frac{A}{2}=1$
   (vi) $\sin A+\sin B+\sin C=4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$
   (vii) $\sin(B+C-A)+\sin(C+A-B)+\sin(A+B-C)=4\sin A\sin B\sin C$.
2. If $A+B+C=2s$, then prove that $\sin(s-A)\sin(s-B)+\sin s\sin(s-C)=\sin A\sin B$.
3. If $x+y+z=xyz$, then prove that $\frac{2x}{1-x^2}+\frac{2y}{1-y^2}+\frac{2z}{1-z^2}=\frac{2x}{1-x^2}\frac{2y}{1-y^2}\frac{2z}{1-z^2}$.
4. If $A+B+C=\frac{\pi}{2}$, prove the following
   (i) $\sin 2A+\sin 2B+\sin 2C=4\cos A\cos B\cos C$
   (ii) $\cos 2A+\cos 2B+\cos 2C=1+4\sin A\sin B\sin C$.
5. If $\triangle ABC$ is a right triangle and if $\angle A=\frac{\pi}{2}$, then prove that
   (i) $\cos^2 B+\cos^2 C=1$
   (ii) $\sin^2 B+\sin^2 C=1$
   (iii) $\cos B-\cos C=-1+2\sqrt{2}\cos\frac{B}{2}\sin\frac{C}{2}$.